UE21CS390A CAPSTONE PROJECT REVIEW #2

(Project Requirements Specification and Literature Survey)

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Project Title:A novel method for video manipulation and compilation using Big Data Technologies

Project ID: 126

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ABSTRACT

- The purpose of this project is to provide a fast and efficient means to process high-quality, next-generation video formats in a parallelized computing form.
- By leveraging Big Data Technologies, the system will address the challenges of handling high bandwidth, large sizes of data, and high speeds of processing.

MOTIVATION AND SCOPE OF THE PROJECT

Motivation

- High demand for top-notch video processing.
- Traditional methods struggle with new video formats.
- Urgent need for faster processing and data handling.

Scope Overview:

- Architecture Development:
  Designing a robust system with Big Data Technologies.
- Algorithm Implementation:
  Creating efficient algorithms for encoding, decoding and editing.
- Hardware Standardization:
  Establishing guidelines for consistent performance.
- Measurable Goals:
  Setting clear targets for speed and efficiency.
- Comprehensive Coverage:
  Addressing all stages from design to optimization.
- Limitations Consideration:
  Identifying and tackling potential challenges swiftly.

USER CLASSES AND CHARACTERISTICS

Educators and Trainers:

- Frequency of Use:Regular usage for creating educational videos and training materials.
- Functionality:Require tools for recording, editing, and sharing instructional content.
- Technical Capability:Possess moderate technical skills, prioritize usability and simplicity.
- Security Levels:Need secure access to educational content and student data.

Streaming Platform Administrators:

- Frequency: Daily use.
- Needs: Tools for managing streaming services.
- Skills: High technical proficiency.
- Security: Require secure access to platform data.

Video Production Professionals:

- Frequency: Regular use.
- Needs: Advanced editing features.
- Skills: High technical proficiency.
- Security: Require secure access to sensitive content.

DEPENDENCIES / ASSUMPTIONS / RISKS

Dependencies:

- Hardware capabilities dictate system design and performance.
- Simulation programs influence development and validation.
- Resource availability affects system scalability and project timelines.

Risks:

- Technology failure: Potential risks of software bugs or system crashes.
- Hardware failure: Risks associated with hardware malfunctions affecting project advancement.
- Version compatibility problems: Risks related to software incompatibilities.

Assumptions:

- Assume access to ample computing resources.
- Assume high quality and compatibility of input video data.
- Assume proficiency of the team in implementing complex algorithms.

FUNCTIONAL REQUIREMENTS

Fundamental Actions:

- Input Validation:
- Validate input videos for compatibility and metadata completeness.
- Ensure codec compatibility before processing.

Data Processing:

- Ingest input video data for pre-processing tasks.
- Utilize distributed computing frameworks like Apache Spark for parallelized computation.
- Ensure fault tolerance and continuous processing.

Workload Distribution:

- Distribute processing tasks across a cluster of machines.
- Manage video processing pipelines with defined phases or stages.

FUNCTIONAL REQUIREMENTS

Error Handling and Recovery:

- Implement error handling mechanisms for unexpected failures.
- Log errors and monitor processing activities for troubleshooting.
- Provide interfaces for manual intervention or recovery.

Parameter Adjustment:

- Optimize resource utilization by adjusting parameters like memory allocation.
- Incorporate performance metrics for evaluation.

Output Generation:

- Generate modified or transformed video outputs based on processing tasks.
- Reassociate metadata and necessary data for output videos.

NON-FUNCTIONAL REQUIREMENTS

Performance Requirement:

- Efficient processing speed for video manipulation tasks.
- Seamless scalability to accommodate growing demands.
- Optimal resource utilization to minimize latency during editing operations.

Safety Requirements:

- Safeguards against accidental deletion or corruption of original video files.
- Robust error handling mechanisms to minimize data loss and ensure system stability.

Security Requirements:

- Role-based access control and robust password policies to limit unauthorized access.
- Secure authentication methods, such as multi-factor authentication, for user identity verification.

LITERATURE SURVEY

Paper Title	Objective	Advantages	Limitations
Big data analytics for MOOC video watching behavior based on Spark Hui Hu, Guofeng Zhang, Wanlin Gao ,Minjuan Wang https://link.springer.com/article/10.1007/s00521-018-03983-z	Analyze student video watching behavior in MOOCs to evaluate effectiveness	Utilizes Spark to analyze large-scale MOOC log data. Offering insights into granular video interactions and determining the relative importance of different behaviors.	The analysis primarily focuses on video behavior, neglecting other aspects of MOOC enga gement. Technical complexity may pose challenges for educators or MOOC owners without a strong data science background

LITERATURE SURVEY

Paper Title	Objective	Advantages	Limitations
Real-Time Big Data Delivery in Wireless Networks Xu Zheng, Zhipeng Cai* https://ieeexplore.ieee.org/abstract/document/7874183	Efficient video delivery in wireless networks	BOSR increases capacity compared to traditional scheduling methods. Reduces rebuffering events. Has potential for adaptation to scenarios with heterogeneous channel conditions.	The computational complexity of finding the optimal scheduling solution is high. BOSR's performance can be sensitive to parameter tuning, requiring careful adjustment for real-world scenarios.

LITERATURE SURVEY

Paper Title	Objective	Advantages	Limitations
Video Transformation in Big Video Era and Its Impact on Content Editing Mingzhi Yin https://www.scirp.org/journal/paperinformation?paperid=113330	Explore the way video content is changing in the era of big video and the impact this has on video editing.	Increased public participation. Emergence of new video forms. Expansion of video communication contexts	Spread of misinformation, Homogenization of video content

LITERATURE SURVEY

Paper Title	Objective	Advantages	Limitations
Scanner: Efficient Video Analysis at Scale Alex Poms, Will Crichton, Pat Hanrahan, Kayvon Fatahalian https://arxiv.org/abs/1805.07339	Large-scale video analysis	Optimized data management. Flexibility Scalability	Complexity in handling temporal dependencies. Maintaining context across multiple frames in a video can be computationally demanding, impacting efficiency.

HYPOTHESIS

Employing Spark-based algorithms for analyzing MOOC video watching behavior improves the understanding of student engagement and aids in the enhancement of MOOC platforms.

The BOSR method improves real-time big data delivery in wireless networks, particularly for video streaming, by optimizing resource allocation and reducing rebuffering events.

ALTERNATE HYPOTHESIS

The Spark-based analysis may not significantly enhance the understanding of MOOC video watching behavior compared to traditional methods, or it may present challenges in implementation and interpretation.

BOSR may not provide significant improvements in real-time big data delivery compared to existing methods, or its effectiveness may be limited under certain network conditions.

HYPOTHESIS

The Big Video Era has reshaped content editing techniques, necessitating adaptations to meet the demands of platforms like TikTok and addressing challenges such as content homogenization.

Scanner provides efficient and scalable video analysis solutions, improving processing times and resource utilization compared to existing methods.

ALTERNATE HYPOTHESIS

The impact of the Big Video Era on content editing techniques may be overstated, or traditional editing principles may remain relevant despite technological advancements.

Scanner's efficiency gains may be marginal or dependent on specific use cases, and its complexity could hinder widespread adoption or implementation.

Discussion

Strengths:

- Efficient use of advanced techniques for data processing and analysis.
- Introduction of novel methodologies and algorithms for video processing.
- Emphasis on optimizing video delivery and enhancing user experience.
- Evaluation through simulations or experiments to validate proposed solutions.

Weaknesses:

- Reliance on limited datasets may hinder generalization of findings.
- Technical complexities in algorithm implementations may pose comprehension challenges.
- Computational complexity and sensitivity to parameter tuning in some approaches.
- Scope limitations and requirement for up-to-date analysis due to rapid technological evolution.

Discussion

Relevant Similarities:

- Utilization of cutting-edge technologies such as big data analytics and real-time processing.
- Focus on addressing challenges in video processing, delivery, and content editing.
- Evaluation methodologies involving simulations or experiments for validation.

Relevant Differences:

- Varied focus areas such as MOOC behavior analysis, wireless network delivery, content editing impacts, and video analysis systems.
- Differences in methodologies, datasets, and technical complexities across studies/products.

Summary of Literature Survey

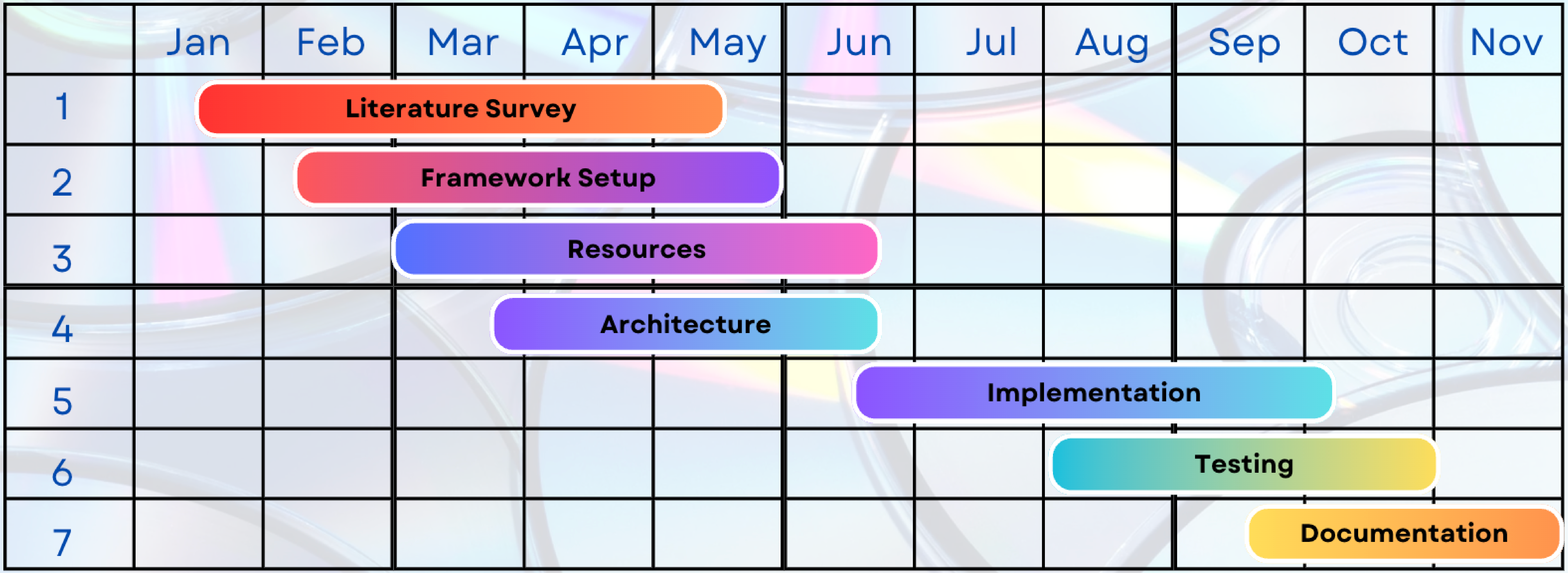
- Utilization of advanced techniques such as big data analytics and real-time processing to address challenges in video manipulation and delivery.

- Insights into user behavior analysis within MOOCs, optimization of video delivery in wireless networks, impacts of the "Big Video Era" on content editing, and development of efficient video analysis systems.

- Strengths lie in the innovative methodologies proposed, such as Spark-based algorithms, BOSR scheduling policy, and Scanner for large-scale video analysis.

- Weaknesses include reliance on limited datasets, technical complexities, and scope limitations in some studies.

Capstone (Phase-I & Phase-II) Project Timeline



Conclusion

- Advanced methodologies in video processing are crucial for addressing real-world challenges.
- Findings highlight the effectiveness of techniques like big data analytics and real-time processing.
- The importance of these findings lies in advancing video processing technologies.
- Future research should refine methodologies for comprehensive video processing solutions.
- Insights gleaned pave the way for advancements with implications across industries.

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Thank you!